

Polymer-Tethered Ligand-Receptor Interactions between Surfaces II

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In this paper, we analyze the thermodynamics of interactions between surfaces mediated by polymer-tethered ligand-receptor binding. From statistical thermodynamics calculations, we obtain an effective two-dimensional binding constant reflecting contributions from the microscopic binding affinity as well as from the conformation entropy of the polymer tethers. The total interaction is a result of the interplay between attractive binding and repulsion due to confinement of the polymer chains. We illustrate the differences between three scenarios when the binding molecules are (1) immobile, (2) mobile with a fixed density, and (3) mobile with a fixed chemical potential (connected to a reservoir), which correspond to different biological or experimental situations. The key features of interactions, including the range of adhesion (onset of binding) and the equilibrium separation, can be obtained from scaling analysis and are verified by numerical solutions. In addition, we also extend our method of treating the quenched case with immobile ligands and receptors developed in a previous paper [Martin, J. I.; Zhang, C.-Z.; Wang, Z.-G. *J. Polym. Sci., Part B: Polym. Phys.* **2006**, *44*, 2621–2637] as a density expansion in terms of both ligand and receptor densities. Finally, we examine several cases having ligand-receptor pairs with different tether lengths and binding affinities, and/or nonbinding linear polymers as steric repellers. Such systems can exhibit complex interactions such as a double-well potential, or a bound state with an adjustable barrier (due to the repellers), which have both biological and bioengineering relevance.

I. Introduction

Cells communicate via ligand-receptor interactions.^{1,2} Such noncovalent interactions, which are present between specific pairs of residues in proteins or polypeptides, are specific (one-to-one) and reversible.³ The interplay between specific and generic physical interactions, such as electrostatic, hydrophobic, and steric interactions,⁴ is crucial to the adhesion and signaling between cells and the extracellular matrix, and it has been extensively studied by researchers in physiology, biochemistry, biophysics, and bioengineering.^{1,2,5–10} Understanding specific cellular interactions, especially their interplay with other generic interactions in biological processes, assists bioengineering design, such as tissue engineering and biospecific recognition. On the other hand, artificially designed biomimetic materials, such as polymersomes,^{11–14} vesicles or liposomes,^{10,15} and substrate-supported monolayer and bilayer membranes,^{16,17} allow better characterization of the

specific and nonspecific interactions because of the absence of complicating factors such as cell signaling or deformation of cell shapes *in vivo*.^{18–22}

Designing materials with biocompatibility requires thorough understanding of the ligand-receptor interactions. In the classical model of cell adhesion proposed by Bell, Dembo, and Bongrand^{23–26} (illustrated in Figure 1), ligand-receptor binding is treated as a chemical equilibrium between ligand and receptor molecules, with the interplay between specific binding and generic physical interactions resulting in an equilibrium constant dependent on the separation between the adhesion surfaces. The Bell model captures the qualitative features of cell adhesion, and it has been successful in fitting certain experimental measurements quantitatively. However, careful inspection of the theory reveals several flaws and confusions. First, the equilibrium constant in the Bell model is by definition for a chemical reaction in two dimensions (2D), which can only be inferred from measurements in bulk solutions (3D), but the relation between these two equilibrium constants is obscure and often causes confusion.^{7,8,19} A rigorous treatment of the statistical thermodynamics of binding in a 2D system is still lacking. Second, a chemical-equilibrium treatment implicitly assumes that the ligand and receptor molecules are mobile on the surfaces, which is valid only when molecules are embedded in a fluid bilayer or membrane. In many experimental settings, ligands and receptors are fixed on beads^{21,22} or covalently linked,^{12,13} in this scenario, the density of bound

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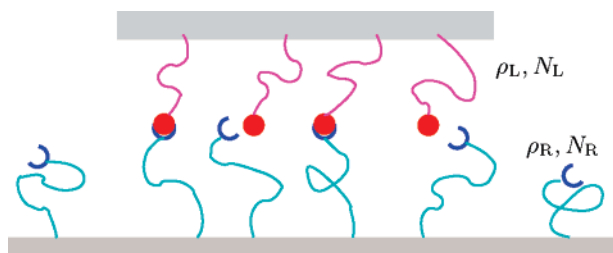


Figure 1. Schematic view of the surfaces with tethered ligands and receptors. The surfaces are separated by a distance L , and the polymer tethers have mean square end-to-end distances $N_L b^2$ and $N_R b^2$, with area densities ρ_L and ρ_R for ligands and receptors, respectively; the anchoring ends are located within a distance z_0 from the surface. In the Bell model, binding between ligands and receptors is treated as a chemical equilibrium with constant K dependent on the surface separation. Here, we assume the binding between a ligand and a receptor residue to have an equilibrium constant K_0 which can be measured in a bulk solution.

ligand-receptor pairs does not follow the mass-action law as is usually implied from chemical equilibrium,²⁷ and it is erroneous to extract the parameters of the Bell model from these measurements by fitting to a chemical-equilibrium expression naively.

In many biological or engineering systems, ligand or receptor groups are tethered by polymers or polypeptides to enhance specificity^{28,29} or achieve different functions.^{30,31} Polymer tethering has also been a common motif in surface force measurements and single molecule studies.^{32,33} The polymer tether turns the short-range, lock-and-key type interaction into a long-range, specific interaction, which has important implications to the equilibrium as well as the dynamic properties of adhesion,^{27,34–36} and suggests a new route to controlling the interactions between surfaces typically achieved by generic physical interactions.^{4,37–39} To characterize the polymer-tethered ligand-receptor binding, we need to separate the contributions from molecular binding and from tether conformation degrees of freedom—a single phenomenological binding constant is inadequate.

In this paper, we study a microscopic model of polymer-tethered ligand-receptor binding and analyze the thermodynamics of binding as well as the interactions between surfaces mediated by the receptors. We explicitly account for the degrees of freedom of the flexible polymer tether and separate their contributions to the effective binding affinity as measured in experiments. Specific attention is paid to the quenched case, where both ligands and receptors are immobile with random distributions. In this scenario, the physical free energy is the average over the random distributions of ligands and receptors (“quenched average”), and an “effective binding constant” that relates the density of bound molecules to the densities of receptors and ligands is not applicable. In the low-density regime, the quenched system has qualitatively different thermodynamics from the annealed system.

We develop an asymptotic expansion of the quenched free energy of binding in terms of the scaled molecular densities of ligands and receptors, which extends our previous analysis for the single ligand problem.²⁷ The leading-order contributions (which are accurate at low densities and intermediate binding strengths) allow us to derive the dependence of the binding free energy and the density of bound molecules on the microscopic binding affinity and the tether chain lengths, which is qualitatively different from results for the annealed systems.

Another feature of this paper is that we distinguish between ligands or receptors with a fixed density (closed), and connected to a reservoir with a fixed chemical potential (open). In the Bell model, the densities of ligands and receptors are both assumed to be the bulk average. However, both experiments¹⁹ and theoretical estimations^{40,41} suggest that, in the adhesion of cells or vesicles, the small contact adhesion zones have ligand-receptor bonds enhanced to much higher densities than the bulk average. In this scenario, the whole noncontact area serves as a reservoir for the small adhesion zones; therefore, the adhesion part should be more naturally treated as an open system with a reservoir of molecules. The biological implications are briefly discussed in section III B. For further discussions in relation to experiments, see ref 41.

We illustrate all our calculations using the ideal-Gaussian-chain model for the polymer tether and highlight the scaling dependence on the contour chain length (Kuhn length); extensions to more complicated realistic models are straightforward.^{29,42} The Gaussian model allows exact analytical calculations of the chain confinement repulsion as well as the chain stretching energy, which are the main features in addition to the specific binding. We find that both at the onset of binding (where ligand-receptor pairs start to form) and at the free energy minimum (bridging conformation is most stable), the surface separations scale linearly as the spatial extension of the Gaussian chain. While the equilibrium separation is found to be insensitive to the binding energy, the onset of binding is proportional to the square root of the binding energy $(\epsilon/k_B T)^{1/2}$ as was suggested in ref 34. These results lead to a quasi-equilibrium critical tension for breaking a tethered ligand-receptor bond that scales as $N^{-1/2}$.

This paper is a sequel to our previous paper²⁷ where a discrete lattice model was used and adhesion was between a single ligand and receptors. In section II, we define the continuum model for tethered ligand-receptor binding as illustrated in Figure 1 and present the theoretical analysis. For simplicity, we choose to study univalent ligands and receptors with monodisperse tether lengths. In section III, we discuss the key features for the simple system corresponding to the model solved in section II, and we go on to study several examples combining different types of specific and nonspecific interactions based on the results in section II. We suggest some tentative guidelines for the control over surface interactions via specific binding. The main conclusions are summarized in section IV with brief discussions on relevant problems.

II. Model and Solution

The model is schematically shown in Figure 1. Definitions of variables are given in Table 1. We assume the anchoring ends on the surfaces to be non-interacting (i.e., ideal gas in 2D), and the surfaces are non-adsorbing and impenetrable for the polymer segments.

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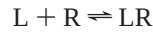
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Table 1. Glossary

variable	definition	dimensions or expressions
$N_i b^2$	mean square end-to-end distance of the polymer tether	$[L]^2$
x_L (x_R)	fraction of tether lengths	$N_L/(N_L + N_R)$
m_i	total number of molecules for each species	
c_i	number of molecules per unit volume	$[L]^{-3}$
ρ_i	number of molecules per unit area	$[L]^{-2}$
ϕ_i	scaled density in 2D	$\rho_i N b^2$
$c_i^{(0)}$, $\rho_i^{(0)}$, $\phi_i^{(0)}$	overall densities (both bound and unbound)	same as above
L	separation between surfaces	$[L]$
l	scaled surface separation	$L/\sqrt{N}b$
l_0 (L_0)	position of the free energy minimum	
l_1 (L_1)	onset of binding	
q_i	partition function of the tethered receptor	
q_i'	internal partition function	$[L]^{-3}$
q_i^t	partition function of the polymer tether	$[L]^3$
F	total free energy	$k_B T$
ΔF	interaction free energy	$k_B T$
F_b , F_r	contribution to ΔF from binding and repulsive confinement	$k_B T$
μ_i	reservoir chemical potential	$k_B T$
Ξ , Q	grand partition function for open systems	
W , p	grand potential, osmotic pressure	$k_B T$
K_0	standard binding constant in terms of c_i	$[L]^3$
K	2D binding constant in terms of ρ_i	$[L]^2$
$(\tilde{\epsilon}) \epsilon$	(effective) binding energy	$k_B T$
A	surface area	$[L]^2$
$A_L^{(0)}$ ($A_R^{(0)}$)	total area of the surface occupied by ligands (receptors)	$[L]^2$
$G(\mathbf{r}, \mathbf{r}'; N)$	Green's function of the polymer chain	$[L]^{-3}$
z_0	anchoring distance	$[L]$
T (T_c)	(critical) quasi-equilibrium tension force to break a single ligand-receptor bond	$[M][L]^{-1}[T]^{-2}$
τ_D , τ_p , τ_r	different time scales	$[T]$

The binding (ligand and receptor) groups are located at the free ends of the polymers. The binding affinity is characterized by the equilibrium constant K_0 of the reaction



in a bulk solution of ligand and receptor proteins (without the polymer tether), as is given by

$$K_0 = \frac{c_{LR}}{c_L c_R}$$

In our calculations, all densities are molecular densities instead of molar densities.

From statistical thermodynamics, we know that

$$K_0 = \frac{q'_{LR}}{q'_L q'_R}$$

where $q'_i = q_i/V$ ($i = L, R$, or LR) is the internal part of the partition function for species i . If molecules are tethered or confined, the translation part is modified, but we may assume that the internal partition function remains the same; that is, q'_i is unaffected by the tether. (Strictly speaking, this assumption is not quite true, since the long polymer tether will affect the binding energy ($k_B T \ln K$) by a few $k_B T$ due to the excluded volume and other local effects. Here, we neglect this effect. Alternatively, we can define K_0 to be the binding constant in a solution of ligands and receptors with polymer tails.) For the current model as illustrated in Figure 1, the equilibrium constant in terms of the surface densities of molecules is given by

$$K = \frac{\rho_{LR}}{\rho_L \rho_R} = \frac{q_{LR}/A}{(q_L/A)(q_R/A)} = \frac{q'_{LR}}{q'_L q'_R} \frac{A q_{LR}^t}{q_L^t q_R^t} = K_0 \frac{A q_{LR}^t}{q_L^t q_R^t} \quad (1)$$

q_i^t includes the conformation as well as the translation degrees of freedom of the polymer tether, which is calculated in section II B.

Throughout the paper we use c_i to denote concentrations in 3D, in units of “number of molecules per unit volume”, and ρ_i for concentrations in 2D (number of molecules per unit area). Later on, we also introduce a dimensionless density ϕ_i which is ρ_i multiplied by the area spanned by the tether.

A. Thermodynamics of the Surface Interactions. We first consider the general thermodynamics for the interactions between surfaces with polymer-tethered ligands and receptors. The total free energy of interaction ΔF is measured by the free energy at a given surface separation relative to infinite separation (non-interacting). Depending on the mobility and the relative fraction of the contact area to the whole surface, each species (ligands or receptors) can be in one of three different scenarios: immobile, mobile with a fixed density, or mobile with a fixed chemical potential (connected to a reservoir). These different scenarios are described by different thermodynamic potentials.

For mobile receptors and ligands, the Helmholtz free energy can be written as

$$F_{\text{tot}} = F(A, m_L, m_R) + F_L(A_L^{(0)} - A, m_L^{(0)} - m_L) + F_R(A_R^{(0)} - A, m_R^{(0)} - m_R)$$

Here, $A_\alpha^{(0)}$ is the total area of the surface occupied by species α , $m_\alpha^{(0)}$ is the total number of molecules, A is the contact area, m_α is the number of molecules within the contact area A , F_α is the free energy of species α in the noncontact region on the surface, and F is the free energy in the contact area.

Before the surfaces are in contact, the free energy is

$$F_{\text{tot}}^{(0)} = F_L(A_L^{(0)}, m_L^{(0)}) + F_R(A_R^{(0)}, m_R^{(0)})$$

Therefore, the net interaction free energy is

$$\begin{aligned}\Delta F = & F_R(A_R^{(0)} - A, m_R^{(0)} - m_R) - F_R(A_R^{(0)}, m_R^{(0)}) \\ & + F_L(A_L^{(0)} - A, m_L^{(0)} - m_L) - F_L(A^{(0)}, m_L^{(0)}) \\ & + F(A, m_L, m_R)\end{aligned}\quad (2)$$

If for both surfaces the entire area is in contact, $A_\alpha^{(0)} = A$, then receptors have fixed overall densities. We have

$$\Delta F = F(A, m_L, m_R) - F_L(A, m_L) - F_R(A, m_R) \quad (3a)$$

If the contact area is a small portion of one surface, for example, $A_L^{(0)} \gg A = A_R$, then the ligands have a fixed chemical potential and the receptors have a fixed density. The free energy of interaction is

$$\begin{aligned}\Delta F = & F(A, m_L, m_R) - F_R(A_R, m_R) \\ & - A \frac{\partial F_L}{\partial A_L} \Big|_{A_L=A_L^{(0)}} - m_L \frac{\partial F_L}{\partial m_L} \Big|_{m_L=m_L^{(0)}} \\ = & F(A, m_L, m_R) - m_L \mu_L - [F_R(A, m_R) - A p_L]\end{aligned}\quad (3b)$$

$p_L = -\partial F_L / \partial A_L$ is the osmotic pressure of the ligands in the reservoir. When the contact area is a small part of both surfaces, both species have a fixed chemical potential, and the interaction free energy is

$$\begin{aligned}\Delta F = & F(A, m_L, m_R) - m_L \mu_L - m_R \mu_R + A(p_L + p_R) \\ = & -A(p - p_L - p_R)\end{aligned}\quad (3c)$$

In these two cases, the appropriate thermodynamic potential is given by $F - \mu_L m_L$ and $F - \mu_L m_L - \mu_R m_R$, corresponding to the cases of open ensemble for ligands (closed ensemble for receptors) and open ensemble for both ligands and receptors, respectively.

If one species is immobile (localized) while the other is mobile, the immobile species essentially corresponds to the closed system with a fixed density. However, if both species are immobile, the translation degrees of freedom are lost. This case is referred to as the “quenched” case. Here, we assume *a priori* that the quenched average free energy is self-averaging, that is

$$\frac{1}{A} \lim_{A \rightarrow \infty} F(A) = \bar{F} = \langle F(\{\mathbf{x}_i\}, \{\mathbf{y}_j\}) \rangle$$

where “ $\langle \rangle$ ” denotes averaging over the position distributions of receptors ($\{\mathbf{x}_i\}$) and ligands ($\{\mathbf{y}_j\}$) and A is the total surface area. The first equality defines the average free energy in the thermodynamic limit, and the second implies that this is equivalent to the average over different distributions of the quenched molecules.

In the quenched case, the binding free energy is a random variable dependent on the distribution of molecules, and its average is essentially different from the annealed cases (where either of the species is mobile). Let us first look at the simple case when ligands and receptors are put on a lattice with no tether, that is, binding occurs only between molecules directly facing each other, then the quenched average of the binding free energy is just

$$\langle F_b \rangle = -\rho_L \rho_R \epsilon$$

where ρ_L and ρ_R are the probability of finding a ligand and receptor molecule within a unit area and ϵ gives the energetic

gain due to binding. Clearly, this is different from the chemical equilibrium in the annealed cases.

Polymer tethers enlarge the range of binding between immobile ligands and receptors, as compared to the molecular case. Nonetheless, at low densities, most molecules are far apart: the physics of interactions is similar to the molecular case with a scaled density. In section II C, we develop an asymptotic expansion for the binding free energy in this low-density limit, from which the density of bound pairs can be obtained.

In the rest of this subsection, we discuss the relevant thermodynamic quantities for each of the annealed cases in detail; these results are quite general and do not depend on the specific model for the tether polymer; explicit treatment of polymer tethering will be described in detail in section II B.

1. *Both-Open System.* First, we examine the system where both receptors and ligands are connected to a reservoir (in a grand canonical ensemble). The free energy of interaction is related to the grand canonical partition function, which is given by

$$\Xi(\mu_L, \mu_R) = \exp[e^{\mu_L/k_B T} q_L + e^{\mu_R/k_B T} q_R + e^{\mu_L + \mu_R/k_B T} q_{LR}] \quad (4)$$

$$= \exp[\lambda_L q_L + \lambda_R q_R + \lambda_L \lambda_R q_{LR}] \quad (5)$$

where $\mu_\alpha = k_B T \ln \lambda_\alpha$ ($\alpha = L, R$) is the chemical potential of the ligands and receptors and q_L , q_R , and q_{LR} are the partition functions of ligands, receptors, and ligand-receptor pairs, respectively.

At equilibrium, the chemical potential of the bound pair μ_{LR} is equal to $\mu_L + \mu_R$; hence, we can rewrite the above equation as

$$\Xi = \exp(\lambda_L q_L + \lambda_R q_R + \lambda_{LR} q_{LR}) \quad (4')$$

And from the grand potential

$$W = -k_B T \ln \Xi = -k_B T (\lambda_L q_L + \lambda_R q_R + \lambda_{LR} q_{LR}) \quad (6)$$

we can obtain the 2D concentrations of ligands, receptors, and bound pairs in equilibrium

$$\rho_i = \frac{1}{A} \frac{\partial \ln \Xi}{\partial \ln \lambda_i} = \frac{q_i}{A} \lambda_i = \frac{q_i^t}{A} \lambda_i q_i' \quad (i = L, R, LR) \quad (7)$$

where q_i' is the internal partition function of the binding group and q_i^t is the partition function of the tether chain including the translation part. From these, we obtain the relation between 2D and 3D binding constant as given by eq 1,

$$K = \frac{\rho_{LR}}{\rho_L \rho_R} = K_0 \frac{A q_{LR}^t}{q_L^t q_R^t}$$

In eq 7, q_i^t depends on the surface separation L . At infinite separation ($L = \infty$), no binding occurs; therefore, $\rho_\alpha(\infty)$ ($\alpha = L, R$) is just the reservoir concentration $\rho_\alpha^{(0)}$,

$$\rho_\alpha^{(0)} = \frac{q_\alpha^t(\infty)}{A} \lambda_\alpha q_\alpha'$$

This relates ρ_i at finite surface separation L to the reservoir concentrations $\rho_\alpha^{(0)}$ as

$$\rho_\alpha = \rho_\alpha^{(0)} \frac{q_\alpha^t(L)}{q_\alpha^t(\infty)}, \quad (\alpha = L, R) \quad (8)$$

$$\rho_{LR} = K \rho_L^{(0)} \rho_R^{(0)} \frac{q_L^t(L) q_R^t(L)}{q_L^t(\infty) q_R^t(\infty)} \quad (9)$$

The interaction free energy per unit area is given by the difference in the grand potential (eq 6)

$$\begin{aligned} \frac{\Delta F(L)}{A k_B T} &= \frac{W(L)}{A k_B T} - \frac{W(\infty)}{A k_B T} \\ &= -\rho_L - \rho_R - \rho_{LR} + \rho_L^{(0)} + \rho_R^{(0)} \\ &= \rho_L^{(0)} \left[1 - \frac{q_L^t(L)}{q_L^t(\infty)} \right] + \rho_R^{(0)} \left[1 - \frac{q_R^t(L)}{q_R^t(\infty)} \right] \\ &\quad - K \rho_L^{(0)} \rho_R^{(0)} \frac{q_L^t(L) q_R^t(L)}{q_L^t(\infty) q_R^t(\infty)} \quad (10) \end{aligned}$$

2. Both-Closed System. In the both-closed system, the total number of molecules are fixed within the contact area. The chemical equilibrium between receptors, ligands, and bound pairs implies

$$K(\rho_L^{(0)} - \rho_{LR})(\rho_R^{(0)} - \rho_{LR}) = \rho_{LR}$$

which gives

$$\rho_{LR} = \frac{1}{2} [\rho_L^{(0)} + \rho_R^{(0)} + K^{-1} - \sqrt{(\rho_L^{(0)} + \rho_R^{(0)} + K^{-1})^2 - 4\rho_L^{(0)} \rho_R^{(0)}}] \quad (11)$$

Here, ρ_{LR} , ρ_L , and ρ_R are 2D concentrations of ligand-receptor pairs, free ligands, and free receptors, respectively, and $\rho_\alpha^{(0)} = \rho_\alpha + \rho_{LR}$ ($\alpha = L, R$) gives the total concentration of ligands and receptors (both free and bound) within the contact area.

The Helmholtz free energy is related to the grand potential by a Legendre transform

$$\begin{aligned} \frac{F}{A k_B T} &= -\frac{W}{A k_B T} + \sum_{\alpha=L,R} \frac{\mu_\alpha}{k_B T} (\rho_\alpha + \rho_{LR}) \\ &= -\rho_L - \rho_R - \rho_{LR} + \sum_{\alpha=L,R} (\rho_\alpha + \rho_{LR}) \ln \frac{\rho_\alpha A}{q_\alpha^t q'_\alpha} \quad (12) \end{aligned}$$

where we have made use of eq 7. (Note that in eq 3a m_α refers to the total number of ligands or species within the contact area, including both free molecules and bound ones. Here, ρ_i refers to the density of the free molecules only.) The interaction free energy is given by

$$\begin{aligned} \frac{\Delta F(L)}{A k_B T} &= \frac{F(L)}{A k_B T} - \frac{F(\infty)}{k_B T} \\ &= \rho_{LR}(L) - \rho_L^{(0)} \ln \frac{q_L^t(L)}{q_L^t(\infty)} - \rho_R^{(0)} \ln \frac{q_R^t(L)}{q_R^t(\infty)} + \\ &\quad \rho_L^{(0)} \ln \frac{\rho_L(L)}{\rho_L^{(0)}} + \rho_R^{(0)} \ln \frac{\rho_R(L)}{\rho_R^{(0)}} \quad (13) \end{aligned}$$

One can verify that if one species is immobile while the other is mobile and both have fixed densities, the only difference is

Table 2. Summary of Different Scenarios

	ligands	receptors	ensemble	expressions
I	immobile	immobile	quenched	25, 28
II	fixed c	fixed c	canonical	3a, 11, 13
III	fixed μ	fixed c	open-closed	3b, 14, 15
IV	fixed μ	fixed μ	grand canonical	3c, 9, 10

the translation entropy of the immobile species, which is independent of binding. Therefore, the free energy of interaction is identical.

3. Open-Closed System. Finally, we examine the case in which one surface has a fixed number of molecules while molecules on the other have a fixed chemical potential. This corresponds to the scenario in which the two surfaces have different overall sizes, for example, a virus binding to a cell, or a vesicle or bead binding to a fluid bilayer.

Assuming that receptors have a fixed overall density, we have

$$\rho_{LR} = K \rho_L (\rho_R^{(0)} - \rho_{LR}) \Rightarrow \rho_{LR} = \frac{K \rho_R^{(0)} \rho_L}{1 + K \rho_L} \quad (14)$$

which is simply the Langmuir isotherm for an ideal gas of ligands. ρ_L is related to the reservoir density as in eq 7

$$\rho_L(L) = \rho_L^{(0)} \frac{q_L^t(L)}{q_L^t(\infty)}$$

The thermodynamic potential of interest is obtained from the grand potential through a Legendre transform over the fixed density (cf. eq 3b), and the free energy of interaction is given by

$$\frac{\Delta F}{A k_B T} = \rho_L^{(0)} - \rho_L(L) + \rho_R^{(0)} \ln \frac{\rho_R(L)}{\rho_R^{(0)}} - \rho_R^{(0)} \ln \frac{q_R^t(L)}{q_R^t(\infty)} \quad (15)$$

This also applies to the case in which receptors are immobile, where the difference due to the translation entropy is a constant independent of binding.

The results are summarized in Table 2. In the next subsection, we study the effects of polymer tethering and derive the expressions of q_i^t for the Gaussian chain model. The quenched case is treated separately in section II C.

B. Polymer-Mediated Specific Interactions. In this subsection, we carry out the exact analytical calculations for polymer-tethered binding. We emphasize that the general theory developed in section II A and in much of section II C and this current section is applicable to any chain model for the polymer tether. However, to provide explicit examples, we illustrate the theory with the Gaussian chain model. A scaling analysis for the general case is given in Appendix A1 in the Supporting Information, and details of the Gaussian chain model and the rigid-rod model are given in Appendices A2 and A3 (Supporting Information).

Since the internal partition function q_i^t is assumed to be unaffected by the polymer tether, all we need is the translation part of the partition function q_i^t modified by the polymer tether. Using the Green's functions of the polymer chain, we can express q_i^t as

$$\begin{aligned} q_{LR}^t &= \int_{\mathbf{r}_L} \int_{\mathbf{r}_R} G(\mathbf{r}, \mathbf{r}_R; N_R) G(\mathbf{r}, \mathbf{r}_L; N_L) \\ &= \int_{\mathbf{r}_L, \mathbf{r}_R} G(\mathbf{r}_L, \mathbf{r}_R; N_L + N_R) \quad (16a) \end{aligned}$$

$$q_\alpha^t = \int_{\mathbf{r}} \int_{\mathbf{r}_\alpha} G(\mathbf{r}, \mathbf{r}_\alpha; N_\alpha) \quad (\alpha = L, R) \quad (16b)$$

Here, $\mathbf{r}$ is the position of the ligand or receptor group in the space between the surfaces, and $\mathbf{r}_L$ and $\mathbf{r}_R$ are the positions of the anchoring ends of ligand or receptor tethers, respectively. Equation 16 applies to any chain model for the polymer tether (as reflected in the Green's functions), as well as to both annealed and quenched cases: for annealed cases, $\mathbf{r}_L$ ($\mathbf{r}_R$) is restricted to the membrane whose integral is over a thin layer within distance z_0 from the surface; for the quenched case, it reduces to a summation over the positions of the immobile molecules.

For ideal Gaussian chains (and some other variants), we can factorize the Green's function,

$$G(\mathbf{r}_1, \mathbf{r}_2; N) = g(\mathbf{u}_1, \mathbf{u}_2; N) h(z_1, z_2; N)$$

where $\mathbf{u}$ and z are the transverse (parallel to the surface) and the longitudinal (perpendicular to the surface) coordinates, respectively, and g and h are the transverse and the longitudinal parts, respectively, of the Green's functions. By translational invariance we have

$$g(\mathbf{u}_1, \mathbf{u}_2; N) = g(\mathbf{u}_1 - \mathbf{u}_2; N)$$

and

$$\int_{\mathbf{u}_1, \mathbf{u}_2} g(\mathbf{u}_1, \mathbf{u}_2; N) = A \int_{\mathbf{u}} g(\mathbf{u}; N)$$

For the end-anchored polymer chain, we further assume that the anchoring distance $z_0 \ll \sqrt{Nb}$, and

$$\int_0^{z_0} h(z, z_1; N) dz \approx z_0 h(z_0, z_1; N) = z_0 h_0(z_1; N)$$

For small enough z_0 , its value does not affect the physical quantities of interest, such as the binding constant or the free energy of interaction.

From eq 1, the binding constant is given by

$$K = K_0 \frac{A q_{LR}^t}{q_L^t q_R^t} = K_0 \frac{h_0(L; N_L + N_R) \int_{\mathbf{u}} g(\mathbf{u}; N_L + N_R)}{\int_{\mathbf{u}} g(\mathbf{u}; N_L) \int_z h_0(z; N_L) \int_{\mathbf{u}} g(\mathbf{u}; N_R) \int_z h_0(z; N_R)} \quad (17)$$

For Gaussian chains, $g(\mathbf{u})$ is a Gaussian distribution and the integration over $\mathbf{u}$ yields unity. We are left with

$$K = K_0 \frac{h_0(L; N_L + N_R)}{\int_z h_0(z; N_L) \int_z h_0(z; N_R)} \quad (18)$$

The one-dimensional Green's function h_0 for a Gaussian chain confined between surfaces can be analytically solved; details are given in Appendix A2 in the Supporting Information.

To highlight the scaling for Gaussian chains, it is convenient to rescale lengths by $\sqrt{Nb}$, which is the mean square end-to-end distance. (Similar rescaling can be done for an arbitrary chain model with $\sqrt{Nb}$ replaced by the mean square chain size.) Here, we choose to scale all lengths by $(N_L + N_R)^{1/2}b = \sqrt{Nb}$. (As is shown in Appendix A2 in the Supporting Information, h_0 is a

scaling function of $L/\sqrt{Nb}$.) With this rescaling, the (dimensionless) receptor/ligand densities are given by

$$\phi_\alpha = \rho_\alpha N b^2 \quad (19)$$

with a dimensionless binding constant

$$\begin{aligned} \frac{\phi_{LR}}{\phi_L \phi_R} &= \frac{K_0}{N b^2} \frac{h_0(L/\sqrt{Nb})}{\int_z h_0(z/\sqrt{N_L}b) \int_z h_0(z/\sqrt{N_R}b)} \\ &= \frac{K_0}{(N b^2)^{3/2}} \frac{\sqrt{Nb} h_0(l)}{q_L^t(l) q_R^t(l)} \end{aligned} \quad (20)$$

(Note that h_0 is of dimension $[\text{length}]^{-1}$; hence, the second factor, which depends on the scaled surface separation $l = L/\sqrt{Nb}$, is dimensionless.)

In the literature, the dissociation constant K_d is frequently reported in units of M (mol/L) and a binding energy ϵ_0 is defined as

$$\epsilon_0 = -k_B T \ln(K_d/[M])$$

which is considered the binding energy measuring the intrinsic binding affinity. For most bound pairs, ϵ_0 is found to be $5-30k_B T$.³⁴ In our problem, K_0 is given in terms of molecular densities, and it is related to K_d as

$$K_0 = K_d^{-1} N_a^{-1}$$

where N_a is Avogadro's number.

Equation 20 suggests that, for tethered binding, we can define an effective binding energy

$$\ln \frac{\phi_{LR}}{\phi_L \phi_R} = \frac{\tilde{\epsilon}(l)}{k_B T} = \frac{\epsilon + \Delta\epsilon(l)}{k_B T} \quad (21)$$

$$\epsilon = k_B T \ln \frac{K_0}{(N b^2)^{3/2}} \quad (22)$$

$$\Delta\epsilon(l) = k_B T \ln \frac{\sqrt{Nb} h_0(l)}{q_L(l) q_R(l)} \quad (23)$$

$\Delta\epsilon(l)$ reflects the energetic cost due to stretching of the polymer tether, while ϵ accounts for the intrinsic binding affinity. ϵ is related to ϵ_0 as

$$\epsilon = \epsilon_0 + \ln \frac{10^{-3} \text{m}^3}{(N b^2)^{3/2} N_a} \quad (24)$$

For tether lengths in the normal range $Nb \approx 10$ nm while $b \approx 0.1$ nm, the second term is of the order of 0.1. Therefore, one can use either ϵ_0 or ϵ as a measure of the binding affinity in the tethered case; we adopt ϵ for convenience in the discussion of different energetic contributions to binding.

C. Immobile Receptors and Ligands: Low-Density Limit.

Since the quenched case (both receptors and ligands are immobile) is qualitatively different from the annealed case, we discuss it in detail in this subsection.

As mentioned above, in this case, the physical quantities of interest are averaged over the quenched distributions of the molecules. We assume the quenched distribution is uniform for each molecule on the surface; that is, the probability that a

particular molecule is found at $\mathbf{r}$ satisfies

$$P(\mathbf{r}_\alpha = \mathbf{r}) = \frac{1}{A} d^2 \mathbf{r}$$

For each species (ligand or receptor), we assume that the distribution of molecules corresponds to a particular realization of the grand canonical distribution controlled by a chemical potential μ (here, we temporarily omit the subscript α for convenience), and then the probability distribution of samples with n molecules in area A is given by

$$P(n) = \frac{1}{Q} \frac{\lambda^n q^n}{n!}$$

where $\lambda = e^{\mu/k_B T}$ and the normalization factor (grand partition function) is

$$Q = \sum_n \frac{\lambda^n q^n}{n!} = \exp(\lambda q)$$

Hence, for the ligand-receptor system, the quenched average free energy is given by

$$\bar{F} = \sum_{m_L, m_R} P(m_L) P(m_R) \langle F(m_L, m_R) \rangle \quad (25)$$

The chemical potential is related to the (thermodynamic average) number density of molecules as

$$\rho_\alpha^{(0)} = \frac{1}{A} \frac{\partial \ln Q}{\partial \ln \lambda_\alpha} = \frac{\lambda_\alpha q_\alpha}{A} \quad (26)$$

Putting this back into eq 25, we have

$$\begin{aligned} \bar{F} &= \frac{1}{e^{A\rho_L^{(0)} + A\rho_R^{(0)}}} \sum_{m_L, m_R} \frac{(A\rho_L^{(0)})^{m_L} (A\rho_R^{(0)})^{m_R}}{m_L! m_R!} \langle F(m_L, m_R) \rangle \\ &= A[\mathcal{F}^{(1,1)} \rho_L^{(0)} \rho_R^{(0)} \\ &\quad + \mathcal{F}^{(1,2)} \rho_L^{(0)} \rho_R^{(0)2} + \mathcal{F}^{(2,1)} \rho_L^{(0)2} \rho_R^{(0)} + \dots] \quad (27) \end{aligned}$$

[Note that we have reserved $\bar{F}$ for the “grand canonical average” and $\langle F(m_L, m_R) \rangle$ can be regarded as the “canonical average” free energy with a given number of molecules, m_L and m_R .] It is easy to see that the largest term in the series in eq 27 has $m_L = A\rho_L^{(0)}$ and $m_R = A\rho_R^{(0)}$; therefore, in the thermodynamic limit, the canonical average $\langle F \rangle$ should be equal to the grand canonical average $\bar{F}$. In Appendix B in the Supporting Information, we present calculations of $\mathcal{F}^{(n,m)}$ up to $n = 2, m = 2$. The density of bound molecules is most conveniently obtained by taking the derivative of F with respect to $\epsilon/k_B T$ (or $\ln K$).

Despite that the expansion is asymptotic, the leading-order results are often qualitatively accurate well beyond the range of densities in which the series is convergent, and here we present the results

$$\bar{F} = \bar{F}_b(\text{binding}) + F_r(\text{repulsion})$$

$$\frac{\bar{F}_b}{Ak_B T} = -\rho_L^{(0)} \rho_R^{(0)} \int_{\mathbf{u}} \ln \left[1 + \frac{3}{2\pi} \exp \left[\frac{\tilde{\epsilon}(l)}{k_B T} - \frac{3u^2}{2Nb^2} \right] \right] \quad (28a)$$

$$\frac{\bar{F}_r}{Ak_B T} = - \left[\rho_L^{(0)} \ln \frac{q_L^t(L)}{q_L^t(\infty)} + \rho_R^{(0)} \ln \frac{q_R^t(L)}{q_R^t(\infty)} \right] \quad (28b)$$

$$\bar{\rho}_{LR} = \frac{2\pi}{3} Nb^2 \ln \left(1 + \frac{3}{2\pi} e^{\tilde{\epsilon}/k_B T} \right) \rho_L^{(0)} \rho_R^{(0)} \quad (28c)$$

III. Results and Discussions

In this section, we discuss the key features of binding between polymer-tethered ligands and receptors, and the overall interactions between surfaces mediated by these polymers. To focus on the key aspects of the problem without complications due to other interactions, we assume the ideal-Gaussian-chain model. Even though the Gaussian chain model is not a faithful representation of real receptors, it serves as a pedagogical example for illustrating the main results of our calculation, and extensions to more realistic chain models are straightforward. Furthermore, we discuss our results, such as the strength and range of binding, in terms of the different length scales based on scaling arguments. Most of these results are rather insensitive to the detailed chain model: the different models only change the scaling dependence of the length scale on the contour length of the polymer tether (cf. Appendix A1 in the Supporting Information).

As we have discussed at the end of section II B, after rescaling by the ideal end-to-end distance of the polymer tether, the dimensionless quantities are listed in Table 3 (cf. Table 1 for definitions of the variables).

Before the discussion, it is informative to estimate the values of these parameters. The average number of ligand/receptor molecules is 10^5 – 10^7 /cell, and the average area of a cell is 10^{-7} – 10^{-6} cm²; these give the average area densities of receptors/ligands $\rho \approx 10^{12}$ /cm². The average tether length (contour length) of integrins and selectins on lymphocyte cells is of the order of 10 nm,³⁰ which is comparable to the estimation in the Bell papers.^{23–25} Assuming the monomer size b to be $\sim 10^{-8}$ cm, we find

$$\phi \approx 10^{12} \cdot 10^{-6} \cdot 10^{-8} = 10^{-2}$$

which gives the overall (dimensionless) density of molecules on the cell surface. In other cases (e.g., a cell adhering to a large surface) where the adhesion zone is a small part of the whole surface (open system), we estimate that $\phi \approx O(1)$ within the contact zone and $\phi \approx 0.001$ outside the contact area (in the reservoir) due to depletion of ligands and receptors.

The binding constant K_0 can be obtained from the dissociation constant K_d . In ref 19, the dissociation constant was found to be $K_d = 6 \mu\text{M}$, and the binding constant is $K_0 = (K_d N_a)^{-1} \approx 10^{-16}$ cm³, which corresponds to a binding energy in our definition

$$\ln \frac{K_0}{(Nb^2)^{3/2}} \approx \ln 10^5 \approx 12$$

in units of $k_B T$. In refs 23–25, the authors estimated the 2D binding constant to be $K^{2D} \approx 10^{-8}$ cm², which gives a binding energy

$$\frac{\epsilon}{k_B T} = \ln \frac{K^{2D}}{(Nb^2)} \approx \ln 10^6 \approx 14$$

in our definition. In ref 34, the authors compiled a list of experimentally measured physical parameters of ligand-receptor binding, and it was quoted that the average binding energy ϵ_0 (cf. section II B) of all available ligand-receptor pairs is about

Table 3. Scaled Variables

surface separation	$l = L/\sqrt{Nb}$
densities	$\phi_i = \rho_i Nb^2$
binding affinity	$\epsilon = k_B T \ln[K_0/(Nb^2)^{3/2}]$
tether fraction	$x_\alpha = N_\alpha/(N_L + N_R)$, $\alpha = L, R$

$15k_B T$. The numerical values of ϵ_0 and ϵ are comparable; in our calculations, we take ϵ to be $10k_B T$ or $15k_B T$.

We proceed as follows in the discussions. In section III A, we study the effects of polymer tether on specific binding and nonspecific repulsion. In particular, we examine the dependence of binding on the tether fractions x_L and x_R . In section III B, we discuss in detail the interactions between surfaces mediated by ligand-receptor binding. We consider the following cases for receptors and ligands: (I) quenched (both ligands and receptors are immobile); (II) “both-closed” (either ligands or receptors are mobile, but both with fixed densities); (III) “open-closed” [ligands are connected to a reservoir, and receptors have a fixed density (either mobile or immobile), or vice versa]; and (IV) “both-open” (both ligands and receptors are connected to a reservoir). Hereafter, we will refer to these different scenarios as “case I” to “case IV”. We focus on the free energy of interaction and the density of bound pairs, as well as on features including the equilibrium separation and the minimum free energy; scaling relations are highlighted and tested by analytical calculations. Finally, in section III C, we study several examples combining different types of specific and nonspecific interactions, including ligand-receptor pairs with different tether lengths or binding affinities, and additional repelling polymers between the surfaces.

A. Effects of the Polymer Tether on Specific Binding and Nonspecific Interactions. From eqs 21 and 28, we see that the binding affinity in both quenched and annealed cases is measured by the separation-dependent effective binding energy

$$\frac{\tilde{\epsilon}(l)}{k_B T} = \frac{\epsilon}{k_B T} + \ln \frac{\sqrt{Nb} h_0(l)}{q_L^t(l) q_R^t(l)}$$

Since ϵ is only weakly dependent on the chain length (cf. Table 3), the dependence of the binding affinity on the surface separation or the tether length is primarily contained in the second term, which gives a stretching energy. In Appendix A2 in the Supporting Information, we have worked out the close-form expressions for the stretching energy in the asymptotic limits of large and small separations

$$\frac{\tilde{\epsilon}(l) - \epsilon}{k_B T} \simeq \begin{cases} \ln \frac{\pi^2}{8l} & (l \ll 1), \\ \ln[3\sqrt{6\pi} l^2 e^{-3l^2/2} \sqrt{x_L x_R}] & (l \gg 1) \end{cases} \quad (29)$$

The effective binding constant (for the annealed case) is given by

$$K = \frac{\rho_{LR}}{\rho_L \rho_R} = Nb^2 e^{\tilde{\epsilon}(l)/k_B T}$$

and from eq 29 this becomes

$$K \propto \begin{cases} \frac{K_0}{l\sqrt{Nb}} = \frac{K_0}{L} & (l \ll 1), \\ \frac{K_0}{\sqrt{Nb}} l^2 e^{-3l^2/2} & (l \gg 1) \end{cases} \quad (30)$$

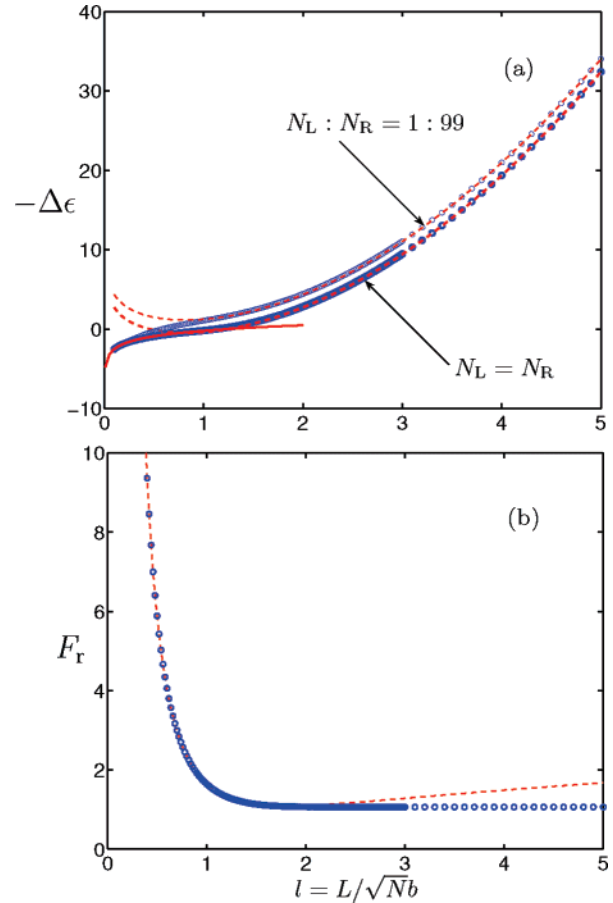


Figure 2. (a) Stretching energy contribution to the effective binding energy ϵ , $-\Delta\epsilon = \epsilon - \tilde{\epsilon}(l)$. Here, $-\Delta\epsilon$ is given in units of $k_B T$ and plotted against the scaled surface separation $l = L/\sqrt{Nb}$. The dashed lines and solid lines are results from the asymptotic expressions in the limits $l \gg 1$ and $l \ll 1$; circles are from exact solutions. The thick dashed line and circles are for equal tether lengths ($x_L = x_R = 0.5$), and the thin line and circles are for a tether length ratio of 1:99. In all calculations, the total tether length $N = N_L + N_R$ is kept constant. (b) Free energy of confining a polymer between parallel surfaces. Circles are results from exact solutions, and the dashed line is from the asymptotic expression for $l \ll 1$.

In refs 23–25, the authors suggested that the 2D binding constant should be related to the 3D binding constant as

$$K = \frac{K_0}{L}$$

Our calculation establishes that this is valid only when surfaces are close enough, that is, the surface separation is less than the ideal size of the tethered ligand-receptor bridge. When surfaces are far separate, the second expression implies a large stretching energy cost. Therefore, one should be careful when inferring a 2D binding constant from the 3D experimental data.

In Figure 2a, we plot the extra stretching energy cost, $-\Delta\epsilon = \epsilon - \tilde{\epsilon}(l)$ in units of $k_B T$, against the scaled surface separation $l = L/\sqrt{Nb}$. We choose two different tether length ratios: the thick lines and circles represent the case with equal tether lengths, $x_L = x_R = 0.5$, while the thin lines and circles are for the case with $x_L = 0.01$ (equivalent to $x_R = 0.01$ by symmetry). Here, the circles represent results from exact solutions, the dashed lines are from the approximate expressions for large separations, and the solid line is from the approximation for small separations (in the latter case, the results only depend on the total tether length and are identical in both cases).

For the entire range of surface separations, the approximate expressions work remarkably well. In particular, the stretching energy cost is given by $3l^2/2$ to the leading order, as was given by scaling arguments.⁴³ We can define the onset of binding l_1 as where $\tilde{\epsilon} \gtrsim 0$, whence the density of bound pairs starts to increase significantly. Assuming that at l_1 the polymer tether is stretched, we can estimate l_1 as (from the asymptotic expression in the strong stretching limit)

$$l_1 = \frac{L_1}{\sqrt{Nb}} \approx \sqrt{\epsilon/k_B T} > 1 \quad (31)$$

by balancing the stretching energy and the binding energy (cf. Appendix A1 in the Supporting Information). Therefore, for $\epsilon \gg k_B T$, $l_1 \gg 1$, the tether chains are considerably stretched when ligand-receptor bridges start to form.

From Figure 2, we also see that, with fixed total tether length $N = N_L + N_R$, binding is optimal if ligand and receptor tether lengths are equal; the difference vanishes at small surface separations as can be inferred from eq 29. The $3k_B T$ difference at intermediate or large surface separations is purely an entropic effect, and it allows fine-tuning of the binding affinity at this range of surface separations without affecting the short-range behavior. This feature is especially relevant near the onset of binding, where $\tilde{\epsilon} \approx 0$: a small difference in $\Delta\epsilon$ can result in a large change in the onset of binding.

Another contribution from the polymers is the repulsion between surfaces due to the confinement of polymer segments, which is shown in Figure 2b. Figure 2b is a scaling plot of the repulsive free energy against the surface separation $l = L/\sqrt{Nb^2}$.

From Appendix A2 (in the Supporting Information), we have obtained

$$\frac{F_r}{k_B T} \approx \begin{cases} -\ln \frac{4z_0}{l\sqrt{Nb}} + \frac{\pi^2}{6l^2} & (l \ll 1), \\ -\ln \frac{4z_0}{\sqrt{Nb}} - \ln \frac{\sqrt{6}}{4\sqrt{\pi}} & (l \gg 1) \end{cases} \quad (32)$$

(z_0 is the distance of the anchoring ends from the impenetrable surface, which does not affect the interaction potential. See Appendix A2 in the Supporting Information for a detailed discussion. In the plot, we have subtracted out the first term in eq 32 involving the anchoring distance z_0 .) The constant in the limit $l \gg 1$ gives the free energy of confining a single Gaussian chain in half space, while the result for $l \ll 1$ scales as $1/l^2$, as is inferred from scaling arguments.

The repulsive free energy ($\sim 1/l^2$) dominates over the binding energy at $l \ll 1$; on the other hand, the stretching energy grows as l^2 at large separations and prohibits binding at large l . Therefore, the net interaction will result in a free energy potential minimized around $l = 1$, where low stretching energy ensures binding and confinement repulsion is small. Next, we discuss the interaction between surfaces mediated by polymer-tethered ligand-receptor binding in different physical scenarios.

B. Interactions between Surfaces Mediated by Ligand-Receptor Binding. In this subsection, we study four different scenarios according to the different mobilities of the molecules: case I (quenched), case II (both-closed), case III (open-closed), and case IV (both-open). While the quenched case applies to interacting surfaces with immobile molecules unambiguously,

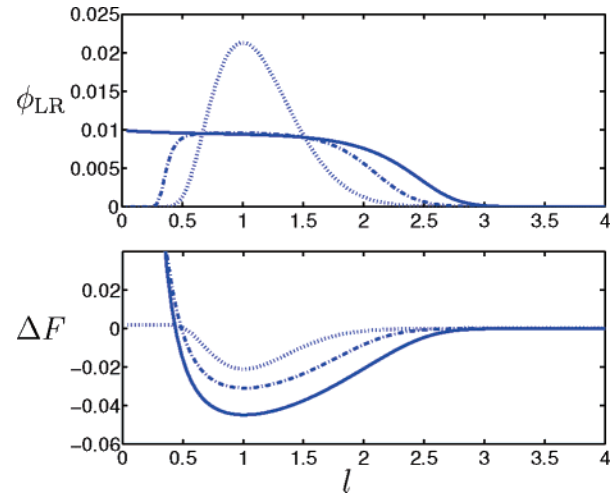


Figure 3. Dependence of the (a) density of bound pairs ϕ_{LR} and (b) interaction free energy ΔF on the scaled surface separation ($l = L/\sqrt{N_L + N_R}b$) in annealed cases (II, III, and IV). The density of receptors or ligands in a closed system is $\phi^{(0)} = \rho^{(0)}(N_L + N_R)b^2 = 0.01$ in case II and case III, and the reservoir density is $\phi^{(0)} = 0.001$ in case III and case IV; the binding energy is $\epsilon = 10k_B T$. Solid lines are results for case II, dash-dotted lines are for case III, and dotted lines are for case IV.

Table 4. Examples of Different Cases

case I (quenched)	polymersomes, solid substrates, colloidal particles with attached polymers
case II (both-closed)	substrate-supporting lipid bilayers and monolayers
case III (open-closed)	flexible membranes interacting with surfaces covered by immobile molecules
case IV (both-open)	flexible membranes, vesicles

the different annealed cases require some comments. First, we note that if one species is mobile, the thermodynamics is the same whether the other species with a fixed density are immobile or mobile. In reality, a closed system (as in case II or case III) is best associated with surfaces with uniformly distributed molecules (mostly artificial surfaces), such as lipid bilayers supported on flat substrates. An open system, on the other hand, corresponds to an inhomogeneous system with a small contact area, such as flexible membranes or lipid bilayers supported on spherical particles. In these cases, the noncontact part of the surface serves as a reservoir for the part in contact. Some typical examples are listed in Table 4.

Here, we focus on two quantities: the density of bound ligand-receptor pairs ϕ_{LR} and the free energy of interactions ΔF . The expressions are given by (cf. Table 2) eq 28 for the quenched case, eqs 11 and 13 for the both-closed system (case II), eqs 14 and 15 for the open-closed system (case III), and eqs 9 and 10 for the both-open system (case IV).

ΔF is the free energy per unit area scaled by Nb^2 . One can rescale ΔF by the overall (scaled) density of receptors, and

$$\Delta F/\phi_R^{(0)}$$

gives the free energy per molecule and coincides with the definition by Bell et al.^{24,25} One can also define the binding fraction

$$f = \phi_{LR}/\phi_R^{(0)}$$

The rescaled ΔF and f allow a comparison with the single-chain calculation in our previous paper.²⁷

(43) de Gennes, P. G. *Scaling Concepts in Polymer Physics*; Cornell University Press: Ithaca, NY, 1979.

First, we compare the different annealed cases (II, III, and IV) where molecules are mobile. In Figure 3, we show results of the scaled density of bound pairs ϕ_{LR} and the interaction free energy $\Delta F(l)$ for these different cases. We choose the receptor density to be $\phi_L^{(0)} = \phi_R^{(0)} = 0.01$ for molecules with fixed densities, as in case II and case III, and we choose the reservoir (of receptors or ligands) density to be $\phi^{(0)} = 0.001$ in case III and case IV. The binding energy is $\epsilon = 10k_B T$; tether lengths are equal for ligands and receptors.

In case II, as the surfaces come close, ϕ_{LR} first increases rapidly and then gradually approaches the overall density $\phi^{(0)} = 0.01$, whence most molecules are bound; in case III and case IV, ϕ_{LR} reaches maximum near $l = 1$. The formation of bound pairs is accompanied by the decrease in the net interaction free energy ΔF . As surfaces come even closer ($l < 1$), the polymers start to feel stronger confinement from the surfaces, which contributes a repulsive potential to ΔF ; when species are connected to a reservoir, the confinement repulsion pushes molecules into the reservoir. The net interaction of repulsive confinement and attractive binding leads to a free energy minimum at $l_0 \approx 1$, and in case III and case IV ϕ_{LR} decreases to 0 at $l \ll 1$ as one or both species are pushed out. In particular, we note that in case IV the free energy flattens off at $F = 0.002k_B T$, which is the total osmotic pressure from the reservoir (equal to the sum of reservoir densities).

In case III and case IV, one or both species are connected to a reservoir; therefore, molecules can diffuse into the contact area from the reservoir when binding is favorable or be pushed out when confinement is strong. The density of bound pairs ϕ_{LR} reflects the flux of molecules accordingly. In case III, we see that even though the reservoir density of receptors $\phi^{(0)} = 0.001$ is much lower than the fixed density of their ligands ($\phi^{(0)} = 0.01$), the maximum density of bound pairs is about 0.01, suggesting that most of the ligands are bound. Therefore, we may conclude that the maximum density of bound pairs is insensitive to the reservoir density of receptors, but depends mostly on the fixed overall density of ligands: this is consistent with the experimental observations in ref 19.

In case IV, even for a low reservoir density of 10^{-3} for both receptors and ligands, the maximum density of bound pairs in the contact area is about $\phi_{LR} = 0.02$, much larger than the reservoir density. This is due to flows of both ligands and receptors into the system, which enhance each other. In particular, the flow of molecules is maximized at optimal binding, and the width of the potential well is narrower compared with those of case II and case III; therefore, the location of the potential minimum is insensitive to the presence of other interactions.

In the surface force measurements by the Israelachvili group,³³ the authors reported experimental results of the bridging force–extension curve. We point out that the bridging force in their setting corresponds to our free energy per unit area. To compare our results with experimental data, a simple conversion is given by (this applies only to the system with a fixed receptor density)

$$F = \Delta F \frac{\rho}{\phi^{(0)}}$$

where ρ is the density of ligands in a real system and $\phi^{(0)}$ is the scaled density we choose. For their density of $\rho \approx 10^{-1}/\text{nm}^2$, this simple estimate gives a maximum bridging force of 2 mN/m (for case II), which is in good agreement with their experimental data. Furthermore, our results suggest that the minimum free energy should be almost independent of the tether length; this is also verified by their experimental results.

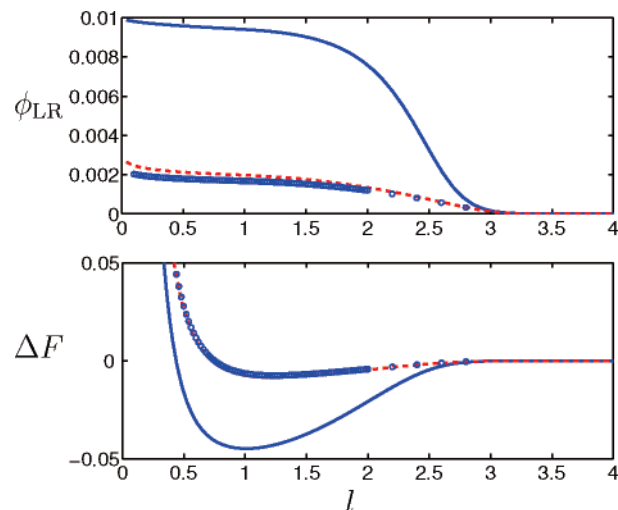


Figure 4. Comparison of case I (“quenched”) and case II (“both-closed”). Solid lines are for case II (mobile ligands and receptors with fixed densities), and dashed lines are for case I (immobile ligands and receptors) from leading-order density expansion; circles are results for the quenched case from density expansions up to quartic order. The binding energy is $\epsilon = 10k_B T$, and both ligands and receptors have a density of $\phi^{(0)} = 0.01$.

Next, we compare the annealed case (we use case II for comparison) and the quenched case (case I). Figure 4 shows ΔF and ϕ_{LR} for case I (“quenched”) and case II (“both-closed”) with a scaled density of $\phi^{(0)} = 0.01$ for both receptors and ligands: the solid lines are results for case II; for case I (“quenched”), the dashed lines are results from the leading-order density expansion $[O(\phi_L \phi_R)]$, and the circles are from expansions up to quartic order $[O(\phi^4)]$. To ensure accuracy of the density expansion, we choose a modest binding energy $\epsilon = 10k_B T$.

In both cases, we see that ϕ_{LR} starts to increase around $L/\sqrt{Nb} = 3.5$, which corresponds to the onset of binding. At the onset of binding, very few bound ligand-receptor pairs are sparsely distributed and the situation is similar to that of isolated non-interacting ligand-receptor pairs; therefore, the scaling dependence of the onset of binding l_1 on ϵ should be identical to that for a single ligand-receptor pair, $l_1 = L_1/\sqrt{Nb} \approx \sqrt{\epsilon/k_B T}$, as discussed in the previous subsection. Indeed, this scaling estimate gives $l_1 \approx 3.2$ for the Gaussian chain model, quite close to the exact results. (Of course, for very strong binding $\epsilon/k_B T \gg 1$, we have $l_1 \gg 1$ and the Gaussian chain model becomes inaccurate. In this regime, the finite extensibility of the polymer chain should be accounted for.)

For the quenched case, we see that ΔF is higher than that in the annealed case and fewer molecules are bound. As discussed in section II A, the free energy in the quenched case is averaged over the random distributions of receptors and ligands. The generic repulsion due to confinement is independent of the positions of ligands and receptors; the quenched average is only invoked for the binding part. By convexity of the free energy,

$$F_q = -k_B T \langle \ln Q \rangle \geq -k_B T \ln \langle Q \rangle = F$$

the quenched average is always larger than the annealed average. Hence, the quenched system always has a higher free energy compared to the annealed case.

At least two factors contribute to the reduced tendency of binding between immobile molecules at low densities. (At high densities, the quenched case should approach the annealed case. See our previous paper.²⁷) First, at low densities, the anchoring ends of ligands and receptors are far apart, so that the tether

chains have to be laterally stretched for ligands and receptors to bind. This lateral stretching adds an extra energetic cost to binding, and it is only dependent on the receptor densities and the tether lengths. Another effect is due to the local inhomogeneity in molecule distributions (this is briefly commented in ref 44): due to fluctuations in the quenched distributions, locally there could be more ligands than receptors or vice versa, and the excess ones have no counterparts nearby and remain unbound, while in the annealed case these molecules can move around to locate their counterparts.

In the Gaussian chain model, the polymer chain is infinitely extensible. We can estimate the average number of molecules within an “accessible” distance as

$$\rho R^2 = \rho N b^2 \epsilon / k_B T = \phi \epsilon / k_B T$$

where the binding energy gain ϵ enables the tether to stretch farther to form a bridge. (When the surface separation is large, ϵ should be replaced by $\tilde{\epsilon}$ to account for the stretching energy.) This gives an estimate for the maximal binding fraction, $f \lesssim \phi \epsilon / k_B T$, which is attained when surfaces are very close. For reasonable values of ϵ with small ϕ , it is always less than unity. (For finitely extensible chains, as in our previous paper,²⁷ there is a strict upper bound set by the average number of molecules within the maximal extension in the limit of $\epsilon / k_B T \rightarrow \infty$, which is completely determined by the molecular density and maximal tether extension.)

We also note that the asymptotic density expansion is only accurate if the surface densities ϕ_L, ϕ_R are small and binding is not too strong. Since the density expansion is carried out around the no binding state, an empirical criterion is given by $\phi \epsilon / k_B T < 1$, corresponding to the “weak” binding scenario. For the parameters we choose, we see that the leading-order expansion is fairly accurate compared to the higher order expansion [up to $O(\phi^4)$]. Since we are mostly interested in the low-density regime when the quenched case and the annealed cases are most different, we shall use the leading-order result throughout the rest of the paper.

From the leading-order expansion, we have

$$\bar{\rho}_{LR} \propto \rho_L^{(0)} \rho_R^{(0)} N b^2 \tilde{\epsilon} / k_B T$$

this should be distinguished from the conventional binding equilibrium,

$$\rho_{LR} \propto (\rho_L^{(0)} - \rho_{LR})(\rho_R^{(0)} - \rho_{LR}) e^{\tilde{\epsilon} / k_B T}$$

In general, there is no well-defined binding constant for the quenched case, especially when higher order terms $O(\phi^3)$ and $O(\phi^4)$ are relevant.

To summarize the binding tendency in different scenarios, we have

$$\text{Both-open} \gg \text{open-close} \gtrsim \text{both-close} > \text{quenched}$$

The difference is due to the entropic effects of flexible polymers as well as the diffusion of molecules in the reservoir. To better illustrate the difference between these cases, we plot in Figure 5 the free energy of binding F_b and the density of bound pairs ϕ_{LR} against the density of one species (receptors or ligands) while keeping the other fixed at $\phi^{(0)} = 0.01$. We see that for the quenched case both ϕ_{LR} and F_b increase almost linearly with the density, and the slope is small ($\propto \tilde{\epsilon}$), reflecting the restricted availability of binding molecules. In case II and case III where the maximum density of bound pairs is set by the fixed density

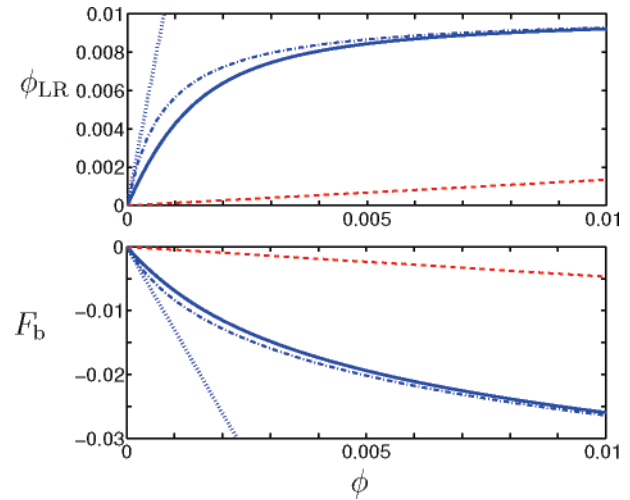


Figure 5. Dependence of the density of bound pairs ϕ_{LR} and the interaction free energy ΔF on the density of molecules for case I (dashed line), II (solid line), III (dash dotted line), and IV (dotted line). The binding energy is $\epsilon = 10k_B T$. In cases I–III, the receptors have a fixed overall density $\phi^{(0)} = 0.01$; in case IV, the receptors have a fixed reservoir density $\phi^{(0)} = 0.01$. The ligand density is varied. Results are calculated for a fixed surface separation $l = 1$ ($L = \sqrt{N}b$), which is near the equilibrium position (cf. Figure 6).

of one species, as the density of the other species increases, ϕ_{LR} first increases exponentially but then approaches saturation. In case IV, the increase is linear all the way with a much larger slope that is proportional to K (cf. eqs 9 and 10); this is in contrast to the quenched case where the slope is proportional to the binding energy $\epsilon \approx \ln K$.

Next, we focus on the features related to the equilibrium separation l_0 and the equilibrium (minimum) free energy. From our scaling analysis in Appendix A1 (in the Supporting Information), for Gaussian chains, the interaction free energy (per bound pair) can be written as

$$\frac{F}{k_B T} \approx -f \left(\frac{\epsilon}{k_B T} - C_1 l^2 \right) + \frac{C_2}{l^2}$$

where $f = \phi_{L,R} / \phi_R$ is the binding fraction and C_1 and C_2 are dimensionless constants. The first term gives the total stretching and binding energy, while the second term measures the repulsion due to confinement. Assuming that f has a weak dependence on l compared to the stretching energy and the confinement repulsion, we obtain the equilibrium separation as

$$l_0 \approx f^{-1/4}$$

and the minimum free energy

$$\frac{F_{\min}}{k_B T} \approx -f \frac{\epsilon}{k_B T} + C_3 f^{1/2}$$

(Indeed, $f \approx 1$ near the equilibrium separation in case II and case III, $\partial f / \partial l \approx 0$ in case IV, and in the quenched case $f \propto \epsilon$ with a logarithmic dependence on l at $l \approx 1$.) The only dependence of l_0 on the binding energy is contained in f . From previous discussions, we have seen that, for a reasonably large binding energy, in the annealed cases, most molecules are bound near the equilibrium position; therefore, in these scenarios, we expect l_0 to reach a constant if $\epsilon / k_B T \gg 1$. This is indeed true as shown in Figure 6.

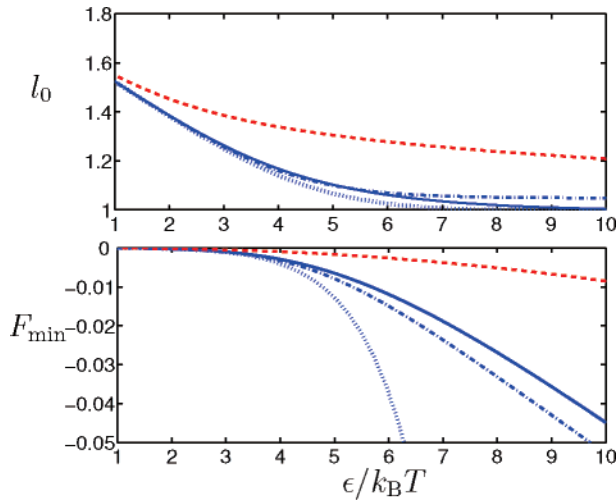


Figure 6. Dependence of the equilibrium separation and the minimum free energy on the molecular binding energy for case I (dashed line), II (solid line), III (dash dotted line), and IV (dotted line). Ligands and receptors have an equal tether length, with densities $\phi^{(0)} = 0.01$.

In Figure 6, we plot l_0 and the equilibrium free energy for different binding energies. The dashed lines represent the quenched case, the solid lines are for case II (both-closed), the dash-dotted lines are for case III (open-closed), and the dotted lines are for case IV (both-open). The receptors and ligands have equal tether lengths, with an equal density $\phi^{(0)} = 0.01$. We observe that, in the annealed cases with strong binding, $l_0 \approx 1$. In the free energy plot, we see that, in case II and case III, the free energy curves approach linear with a slope of 1 for large ϵ , reflecting that in this regime most molecules are bound ($f \approx 1$); in case IV (both-open system), the free energy has an exponential increase due to the incoming molecules from the reservoirs.

On the other hand, the quenched case is essentially different. From above, we have estimated that $f \leq \phi\epsilon/k_B T$ and therefore $f \ll 1$ for a not too large ϵ . A naive estimate for l_0 gives $l_0 \approx f^{-1/4} \approx (\epsilon/k_B T)^{-1/4} > 1$, which would be true only if f is independent of l . Even though the latter assumption does not hold in this regime, the qualitative trends still hold: we indeed observe that l_0 is larger compared to the annealed cases and decreases as ϵ increases.

From the interaction free energy, we can calculate the equilibrium force defined as

$$T = \frac{\partial \Delta F}{\partial L}$$

T measures the force between the surfaces at a given surface separation. The maximum in the force–extension curve corresponds to the critical pulling force above which the bond will be broken even in the quasi-equilibrium state (without fluctuations), which gives an upper bound of the bridging force.³⁵ From scaling arguments, we would expect

$$T_c \sim \frac{\phi_{LR}\epsilon}{(l_1 - l_0)\sqrt{Nb}} \sim \phi_{LR}(\epsilon k_B T)^{1/2} N^{-1/2}$$

for stretched tethers that are still in the Gaussian regime. Our numerical result gives that the breaking force of a single bond is about 10 pN for a tether length of 1 nm and binding energy of $\epsilon = 10k_B T$. This value is comparable to the numerical result reported in ref 35 in the quasi-stationary scenario.

In summary, the interactions between surfaces mediated by polymer-tethered ligands and receptors include the generic repulsion due to confinement and the specific attraction due to binding. The magnitude of the binding attraction is determined by the microscopic binding affinity, the tether lengths, and the molecular densities through an effective binding constant and the scaled molecular densities. The net effect of binding attraction and confinement repulsion results in a free energy minimum at a surface separation comparable to the ideal size of the polymer tether.

When one or both species are connected to a reservoir, molecules can be attracted into or pushed out of the system according to the interaction energy. Qualitatively different is the case when molecules are immobile. The free energy cost due to lateral stretching makes binding much less probable, resulting in a much shallower potential minimum.

We further note that, in all these cases, the onset of binding (where the density of bound pairs starts to increase considerably) appears to be identical, reflecting that the effective binding energy $\tilde{\epsilon}$ is a universal measure of the binding tendency, while the equilibrium bound state, which is dependent on the density of bound molecules, differs in the different scenarios due to entropic effects (diffusion of molecules). The dependences of the equilibrium separation and the minimum free energy can be qualitatively explained by scaling arguments.

Since our results are based on equilibrium analysis but real systems or processes are usually nonequilibrium in nature, it is natural to ask in what situations these conclusions hold. Let us take the surface force measurement^{32,33} as an example. Here, we observe three physical time scales, corresponding to the binding reaction (τ_r), the diffusion of the polymer tether (or the ligand/receptor group) in solution (τ_p), and the transmembrane diffusion of polymers (τ_D). In addition, there is the time scale corresponding to the relative speed at which surfaces are approaching or departing from each other (τ). In ref 34, the authors found that the polymer diffusion time τ_p (Zimm time) is roughly 1 μ s and the binding reaction time τ_r is typically several nanoseconds. The diffusivity of proteins across the membrane was quoted in refs 10 and 19 to be 10^{-8} – 10^{-9} cm²/s, which gives a time scale of $\tau_D \approx (\rho D)^{-1} \approx 1$ ms for the rearrangement of molecular distributions. Therefore, we have

$$\tau_D \gg \tau_p \gg \tau_r$$

Generally, $\tau \gg \tau_r$, and it is always valid to treat the binding reaction as an equilibrium. If the surfaces approach very fast such that $\tau < \tau_p$, then the diffusion of the polymer tether is relevant. This is the scenario analyzed in refs 35 and 44 using reaction–diffusion theory. If the surfaces approach slow enough such that $\tau > \tau_D$, then the system is essentially governed by the equilibrium thermodynamics and all our results should hold; if $\tau_D > \tau > \tau_p$, then the system is in the quenched scenario, as the diffusion of the molecules across the membrane is too slow to be treated as annealed.

In refs 34, 35, and 44, the authors discussed the relevance of the approaching speed to the interacting force between the surfaces and the dependence of the range of binding on the binding energy. For the dependence of the onset of binding (their “binding range”) on the binding affinity, they found the same result as ours, $l_1 \propto \sqrt{\epsilon/k_B T}$, which sets an upper bound in the dynamic measurements where surfaces approach at a finite speed; for the dependence on the tether length, our result suggests $l_1 \approx$

(44) Moreira, A. G.; Jeppesen, C.; Tanaka, F.; Marques, C. M. *Europhys. Lett.* **2003**, *62*, 876–882.

$\sqrt{\epsilon/k_B T N^*} b$, which agrees with their experimental results for long tethers (the authors did not do a scaling plot in refs 44 and 35). For short tethers, the finite extensibility should change the scaling dependence. The breaking of the bond is more subtle and is best put in a dynamic context.^{36,45} However, as discussed in ref 35, the equilibrium force gives an upper bound on the bridging force or the breaking force as long as the approaching or separating speed is not faster than the relaxation of the polymer segments. And our prediction that $T \approx \epsilon^{1/2}/N^*$ should hold for long chains in this regime. Thermal fluctuations will lower the threshold for breaking the bond, as was considered in ref 36.

The distinct time scales of motion result in different physical scenarios. For example, the diffusion of ligand and/or receptor groups is governed by the diffusion of the polymer tether as well as the diffusion of the anchoring end in the membrane. However, the diffusion in the bilayer is much slower compared with the diffusion of polymer segments in solution; therefore, polymer-tethered ligands and receptors can locate their counterparts more easily, resulting in faster adhesion dynamics compared with adhesion without tether, as was observed in ref 15. In addition, the polymer tether increases the range of binding, which admits larger membrane deformations compared to the case of molecular binding: such membrane fluctuations lower the energy barrier of adhesion and stabilize the bound state. These two effects qualitatively explain the experimental findings in ref 15, although quantitative treatments require an analysis of the adhesion dynamics, which is beyond the scope of our current paper.

C. Composite Interaction Potential from Specific Binding and Nonspecific Interactions. In previous subsections, we discuss the interactions between surfaces mediated by polymer-tethered ligand-receptor binding. Real biological processes, however, usually involve many different types of ligand-receptor interactions, with different binding affinities or tether lengths. See ref 30 for a snapshot of different receptors involved in immunological responses. Even in a simple cell adhesion, the polysaccharide layer on the cell surface introduces additional repulsion between the cell surface and the external surface. This repulsive layer effectively prevents nonspecific binding and preserves specific adhesion.

Here, we study the overall interaction potential between surfaces mediated by several specific and nonspecific interactions. We neglect nonspecific intra- and intermolecular interactions, such as the excluded volume, and focus on the features of specific interactions and of the generic repulsion. From these examples, we try to illustrate the different features associated with each interacting species and provide some general principles for the design and control of surface interactions.

1. Cell Adhesion Revisited. Bell, Dembo, and co-workers^{23–25} first proposed that cell adhesion is a net result of specific ligand–receptor binding and nonspecific steric repulsions due to repeller molecules on the cell surface. Here, we re-examine this model and study the dependence of the interaction potential on measurable and controllable molecular parameters, which can guide bioengineering design of artificial surfaces that can trigger cell adhesion. Specifically, we treat the binding molecules as polymer-tethered ligands and receptors and the repelling polymers as linear Gaussian chains confined between surfaces. (Simple scaling tells us that the short-range repulsion due to confinement of Gaussian chains scales as $\sim Nb^2/L^2$. In the Bell model, the repulsion potential is assumed to be $\propto L^{-1}$; this would correspond to stretched polymers in the brush regime. Extension to this scenario can be straightforwardly implemented via a self-

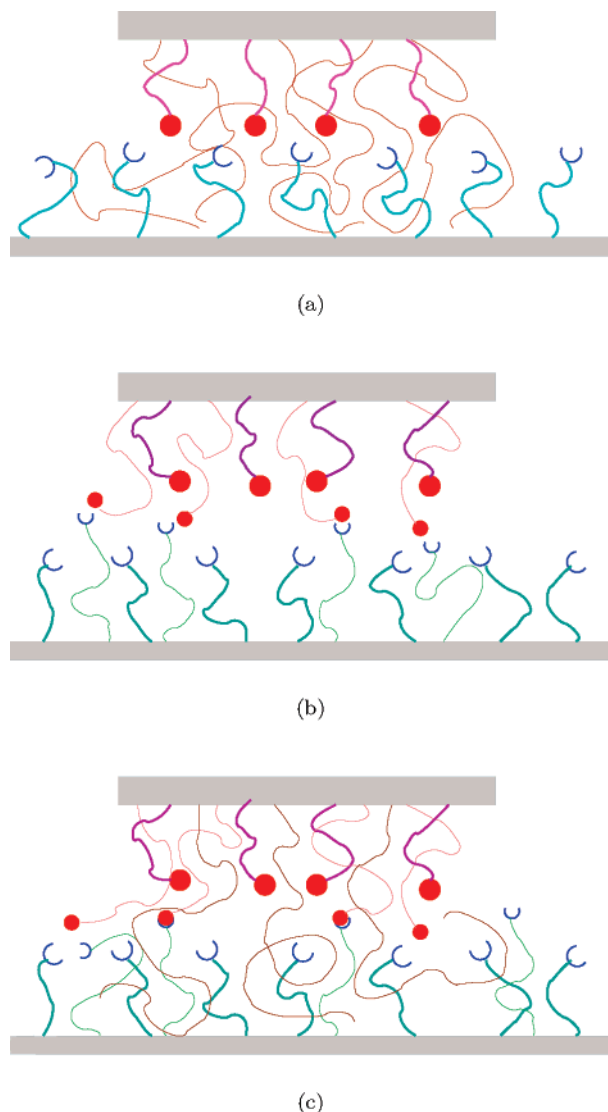


Figure 7. Schematic views of the models discussed in section III C1 (a), section III C2 (b), and section III C3 (c).

consistent calculation.) The model is schematically shown in Figure 7a.

Figure 8 shows the composite interaction potential due to ligand–receptor binding and steric repellers. In Figure 8a, the system belongs to case III (open-closed system) and we plot the examples with mobile repellers (dashed line) and immobile repellers (thick solid line). The interaction potential due to ligand–receptor interactions alone is shown for comparison (thin solid line).

The repeller polymer has length $N_r = 16(N_L + N_R)$, and hence, the repulsion is present at

$$L \approx \sqrt{N_r} b = 4\sqrt{N_L + N_R} b \gtrsim \sqrt{\epsilon/k_B T} \sqrt{N_L + N_R} b$$

which is slightly larger than the onset of binding. (In this subsection, the interaction potential and the separation between surfaces are always scaled by the size of the short tether.) When repellers are mobile, the repulsive potential flattens off at small separations, implying that they are squeezed out (or “redistributed” in Bell’s terminology). This introduces a modest barrier (osmotic pressure) that is proportional to the density of repellers; the length of repellers only affects the range of repulsion but not the barrier height. When repellers are immobile, the confinement repulsion scales as $N_r b^2/L^2$ and presents a strong repulsion. Accordingly,

(45) Evans, E.; Ritchie, K. *Biophys. J.* **1999**, *76*, 2439.

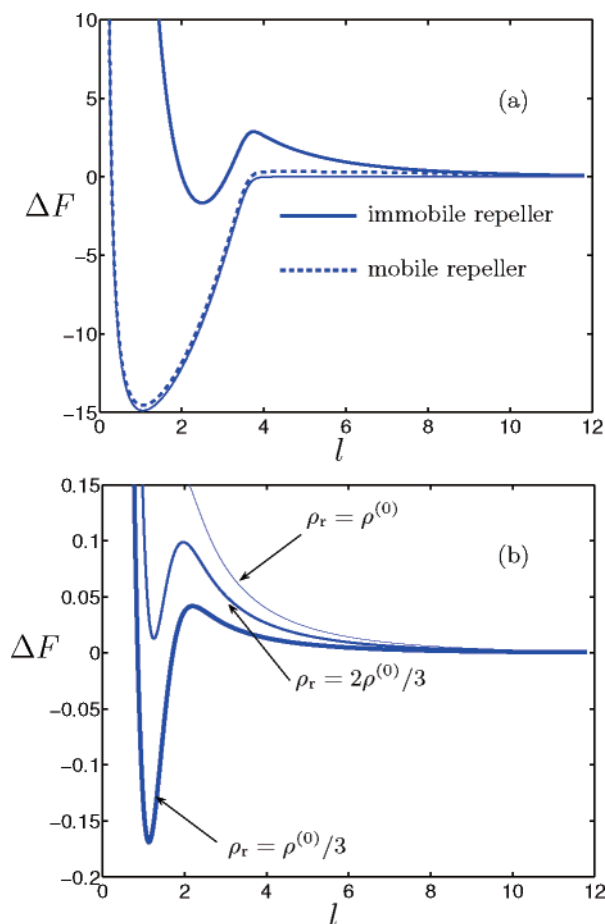


Figure 8. Interaction between surfaces mediated by ligand-receptor binding and repulsive polymers. The binding energy is $\epsilon = 15k_B T$; the length of repelling polymers is $N_r = 16(N_L + N_R)$. In (a), the density of ligands and receptors is $\rho^{(0)}(N_L + N_R)b^2 = 1$ and the repeller density is $\rho_r = \rho^{(0)}/3$. The dashed line is for mobile repellers, the thick solid line is for immobile repellers, and the thin line is for the bare ligand-receptor system without repellers. (b) Receptors and ligands for case IV with a reservoir density $\rho^{(0)}(N_L + N_R)b^2 = 0.005$. The binding energy is $15k_B T$; from above, the densities of repelling polymers are $\rho_r = \rho^{(0)}$ (thinnest), $\rho_r = 2\rho^{(0)}/3$ (moderate), and $\rho_r = \rho^{(0)}/3$ (thick). The length is scaled by the size of the short tether.

the equilibrium bound state is shifted toward larger surface separations with a higher free energy.

In Figure 8b, we examine the case when both receptors and ligands are connected to a reservoir with density $\phi^{(0)} = 0.005$, together with immobile repeller polymers at different densities. Contrary to the case with a fixed overall receptor density, the location of the potential equilibrium is shifted very little, although the magnitude of the potential depth changes. Therefore, the “both-open” system allows us to adjust the free energy of the bound state while keeping its location fixed.

Recently Bruinsma, Sackmann, and co-workers⁴⁰ studied the adhesion between a large vesicle and a lipid bilayer when both receptors and repellers are present. They observed that tightly bound regions with higher densities of receptors coexist with loosely bound states with lower densities and that receptors slowly diffuse to the tightly bound regions (focal adhesion zone). This coexistence was argued to be resulting from a double-well intermembrane potential separated by a barrier induced by the repeller molecules. The authors further pointed out that the repellers are better characterized as mobile with a fixed chemical potential so that they can be pushed out in the tightly bound regions.

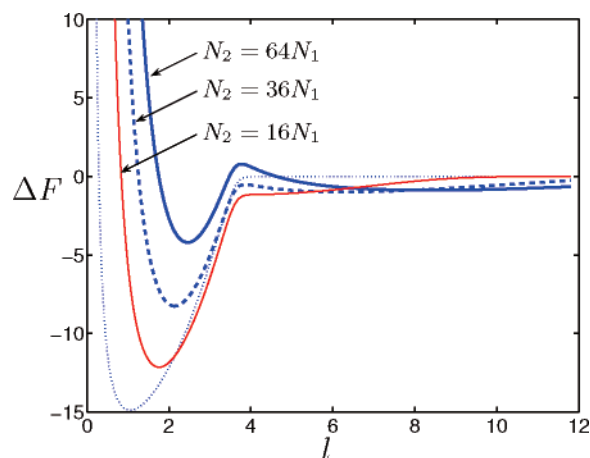


Figure 9. Interaction potential resulting from binary ligand-receptor interactions. Ligands and receptors have equal densities. The short-tethered ligand-receptor pairs have $\epsilon_1 = 15k_B T$ and $\phi_1^{(0)} = 1$, and the system belongs to case III (open-closed). The thin dotted line represents the interaction potential of this system alone. The long-tethered ligand-receptor pairs have a weaker binding energy $\epsilon_2 = 5k_B T$ with a fixed density (case II) $\rho_2^{(0)} = 0.5\rho_1^{(0)}$. Both ligand-receptor pairs have equal lengths for ligands and receptors. From above, the lengths of the longer tethers are $N_2 = 64N_1$ (thick solid line), $N_2 = 36N_1$ (dashed line), and $N_2 = 16N_1$ (thin solid line). The length is scaled by the size of the short tether.

In our model, we do not account for the long-range physical interactions or membrane undulations, which result in the loosely bound state at larger separations as described in the paper by Bruinsma et al. Otherwise, our results on the interaction potential qualitatively agree with their calculations. In addition, we point out that since the growth of the contact area is slow, initially the adhesion zone should be viewed as an open system of receptors with the loosely bound part serving as the reservoir. (This is also observed in ref 19.) The attraction inside the focal contact is significantly higher than that predicted from the overall density on the surface, which can overcome the barrier due to immobile repellers; such a process would be impossible if the binding molecules were uniformly distributed as in a closed ensemble.

We also point out that, due to the barrier to the bound state, even in flat geometries, the adhesion process should be a first-order transition.^{41,46} Therefore, the barrier height controls the dynamics of adhesion, and the presence of a considerable barrier can prevent nonspecific adhesion. In this case, the growth of the adhesion contact is through nucleation, which should be mediated by membrane deformations. We study this interplay between membrane deformations and ligand-receptor interactions in a forthcoming paper (Zhang, C.-Z.; Wang, Z.-G., submitted to *Phys. Rev. E*).

2. Binary Ligand-Receptor Binding. Introducing long repelling polymers can generate a barrier from the unbound to the bound state, thus preventing unwanted adhesion between the surfaces. If, instead of purely repelling polymers, we introduce longer-tethered ligands and receptors, then these molecules act as a “barrier” to the shorter-tethered binding, but on the other hand they also generate another bound state at a larger surface separation. Properly adjusting the binding affinities and tether lengths of these two ligand/receptor pairs gives us extra freedom in controlling the strength and range of the attraction between the surfaces.

In Figure 9, we plot the interaction potential for a system with two types of receptors, as schematically represented in Figure

(46) Weikl, T. R.; Andelman, D.; Komura, S.; Lipowsky, R. *Eur. Phys. J. E* **2002**, *8*, 59–66.

7b. The shorter-tethered receptors have a larger binding energy $\epsilon = 15k_B T$ and higher density $\phi_1^{(0)} = 1$, and we assume them to be an “open-closed” system (case III) to mimic cell–substrate interactions. The longer-tethered receptors have a smaller binding energy $\epsilon = 5k_B T$ and smaller density $\rho_2^{(0)} = 0.5\rho_1^{(0)}$ (note that this is the molecular density instead of the scaled density ϕ), and we assume the longer-tethered molecules to be in a closed system (case II).

Since the long tethers introduce a barrier to the short-tethered binding and generate a new free energy minimum at larger separation, the superimposed interaction potential should have a double-well shape. In Figure 9, the thick solid line represents the case with tether length ratio $N_2/N_1 = 64$, the dashed line is for $N_2/N_1 = 36$, and the thin solid line is for $N_2/N_1 = 16$; the dotted line is for the system with short-tethered receptors only (the same as in Figure 8a). For the range of tether lengths we studied, we see that the minimum due to the short-tethered binding is shifted to larger separations with higher free energies, similar to that in Figure 8a due to repellers.

When the length of the long tether is much larger than that of the short tether (thick solid line), we observe two minima separated by a positive barrier. If the long tether is of intermediate size (dashed line), we still observe two separate minima, but the barrier between them is small; for comparable tether lengths (thin solid line), the two minima merge into one with a larger range of attraction. These results demonstrate that by adjusting the relative length ratio one can qualitatively control the shape of the interaction potential from a single-well to double-well shape.

Understanding the interactions mediated by binary ligand–receptor binding is relevant to both surface engineering and our understanding of biological systems. In a colloidal suspension with particles interacting via a double-well potential, we expect structures with complicated symmetries as a result of ordering at competing length scales, as well as colloidal gels with local but no long range order. (See ref 37 for some examples using single ligand–receptor pairs.) On the other hand, it is known that, in the rolling of leukocyte cells,^{18,31} the longer but weaker selectin ligands mediate the rolling of cells, while the shorter but stronger integrin receptors result in the final strong adhesion; the interplay between longer-tethered receptors and shorter-tethered receptors might be crucial for stable rolling, which is a prerequisite for immunological responses.

3. Attempt at a Synthesis. From the above examples, we have seen two ways to adjust the interaction potential between surfaces: (1) introduce a barrier to the bound state by adding long repelling polymers and (2) shift the equilibrium separation and the minimum free energy and allow different minima to appear by combining ligand–receptor pairs of different lengths.

We now summarize the main features in each case. To introduce a barrier to the bound state, mobile repellers introduce a less noticeable barrier and a smaller shift to the equilibrium separation compared to immobile repellers; for immobile repellers, increasing the density or the chain length of the repeller molecules can increase the barrier height, but the latter will also result in a larger range of repulsion. When a different type of ligand/receptor pair with longer tethers is introduced, depending on the tether lengths, the system can show two free energy minima separated by a large barrier (large tether length difference), two minima separated by a small barrier (intermediate tether length difference), or one minimum with a larger range of attraction (comparable tether lengths). Adjusting the density of each type of ligand/receptor molecule allows us to control the relative stability of the minimum due to each ligand–receptor pair.

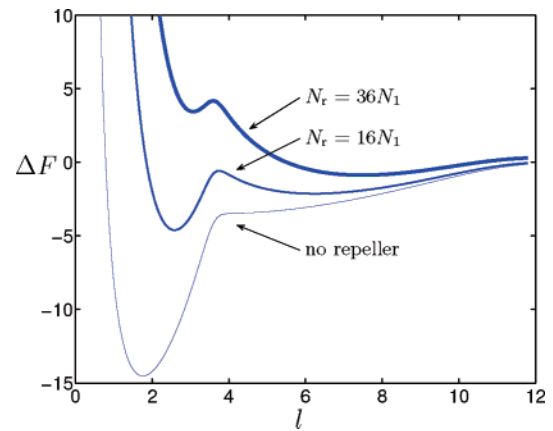


Figure 10. Interaction between surfaces with two different ligand–receptor interactions and repelling polymers. The short-tethered receptors (case III) have $\epsilon_1 = 15k_B T$ and $\phi_1^{(0)} = 1$; the long-tethered receptors (case II) have $\epsilon_2 = 10k_B T$, $\rho_2^{(0)} = 0.5\rho_1^{(0)}$, and $N_2 = 16N_1$. Ligand and receptor tether lengths are equal for both types. The lengths of immobile repelling polymers are $N_r = 36N_1$ (thickest line) and $N_r = 16N_1$ (intermediate) with density $\rho_r = \rho_1^{(0)}/3$. The length is scaled by the size of the short tether.

Combining these two methods allows us to control the subtle features of the interaction potential. Here, we just give one example to illustrate how the relative stability of the minima in a binary system can be controlled by introducing repellers of different lengths. In Figure 10, the light thin line represents the system with two kinds of ligand/receptor pairs with parameters $\epsilon_1 = 15k_B T$, $\phi_1^{(0)} = 1$ and $\epsilon_2 = 10k_B T$, $N_2 = 16N_1$, $\rho_2^{(0)} = 0.5\rho_1^{(0)}$. Comparing Figure 10 with Figure 9, we see that although the binding energy for the longer-tethered ligand/receptor molecules is larger, the qualitative features are identical; hence, changing the binding energy has little effect on the shape of the interaction potential. However, by introducing immobile repeller molecules with a fixed density $\rho_r = \rho_1^{(0)}/3$ but different lengths, we can qualitatively control the interaction potential. We observe two separate minima when repellers are present. For long repellers ($N_r = 36N_1$, thickest line), the stable one is at the larger separation, corresponding to the longer-tethered binding, and the repellers generate a barrier to the bound state. In the case of short repellers ($N_r = 16N_1$), the stable bound state is at the smaller separation and there is no barrier from the unbound to the longer-tethered bound state.

Clearly, one can introduce more species into the system to adjust individual features independently. Because of the specificity of ligand–receptor binding, these different interactions are independent, and the total interaction is the superposition of all ligand–receptor interactions: this provides a diverse and powerful way for engineering surface interactions.

IV. Conclusion

We have studied a continuum microscopic model for polymer-tethered ligand–receptor interactions between surfaces and analyzed the thermodynamics of interactions between the surfaces, which essentially consist of a repulsion due to the confinement of polymers at small surface separations and an attractive binding at an intermediate range of separations mediated by the polymer tether. The generic short-range repulsion due to confinement can be calculated or estimated from scaling analysis for a given chain model. For the tethered binding, we find an effective binding constant that relates the density of bound pairs to those of ligands and receptors at a given surface separation.

The binding constant has contributions from a microscopic binding affinity between the ligand and the receptor group, which is independent of the surface separation or tether lengths, and a tether stretching energy, which reflects the conformation change of the polymer tether due to binding. At small surface separations, the 2D binding constant is independent of the tether length and can be related to the 3D binding constant by $K^{2D} = K^{3D}/\text{surface separation}$, but at large surface separations the stretching energy is important. The attractive binding and the repulsion due to confinement result in an equilibrium separation between the surfaces corresponding to the minimum of the total interaction free energy.

For the overall interactions between surfaces, we study the different scenarios when binding molecules have different mobilities. Specifically, ligands and receptors can be immobile, mobile with a fixed density, or mobile with a fixed chemical potential. These different scenarios correspond to binding between objects with different geometries or different molecular embeddings. Binding is least probable in the case when both species are immobile. On the other hand, in the cases with open ensembles, molecules are attracted into or pushed out of the system due to the net interaction, resulting in a lower free energy. In particular, for the case with ligands in a closed system and receptors in an open system, the maximum density of bound ligand-receptor pairs is determined by the fixed density of ligands, and it is insensitive to the reservoir density of receptors, as is observed in experiments.

We illustrate our calculations using an ideal-Gaussian-chain model. Simple scaling arguments suggest that the onset of binding

(adhesion range) scales as $L_1 \approx \sqrt{\epsilon/k_B T} \sqrt{Nb}$ and that the equilibrium separation scales as $L_0 \approx \sqrt{Nb}$. These results agree well with the exact solutions. We also infer that the quasi-equilibrium critical tension for breaking a ligand-receptor bond should scale as $N^{-1/2}$ for the Gaussian tether. These scaling relations should also hold for non-Gaussian chain models by replacing the $N^{1/2}$ factor with the characteristic size of any polymer chain model, $R \approx N^\nu$.

Finally, we demonstrate that, by combining different types of ligand-receptor interactions and nonspecific repeller molecules, one can achieve precise control over the interaction potential between surfaces. Specific examples include introducing a barrier to the bound state and introducing multiple minima and controlling the range and the depth of each minimum. These results suggest possible strategies for bioengineering design with better specificity and for a diverse control of surface interactions using ligand-receptor interactions.

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Supporting Information Available: Scaling analysis (Appendix A1) of the key features of surface interactions due to ligand-receptor binding, including the range and the strength of binding, for both Gaussian chains and the self-avoiding chains; details of analytical calculations of the Green's functions for Gaussian chains (Appendix A2) and the rigid-rod model (Appendix A3); and perturbative calculation of the quenched free energy (Appendix B). This material is available free of charge via the Internet at <http://pubs.acs.org>.

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